

PLC Tabletop Training Program

- Detailed Training Guide & Technical Reference
- April 2026

Module 0: Electrical Safety & LOTO

- Personnel safety fundamentals required before any hands-on work. Covers lockout/tagout, arc flash awareness, and safe work practices.
- Describe LOTO procedures (OSHA 29 CFR 1910.147) for PLC cabinets
- Identify when energized troubleshooting is permitted vs. when LOTO is required
- Understand arc flash PPE categories for PLC cabinet voltages
- Demonstrate safe multimeter use and proper CAT-rated equipment

Module 1: PLC Hardware & Components

- Comprehensive walkthrough of PLC cabinet components, I/O modules, power distribution, and cabinet infrastructure.
- Identify every major component inside a PLC control cabinet from physical inspection
- Explain the function of the CPU, power supply, Ethernet board, I/O modules, and terminal blocks
- Distinguish between digital I/O and analog I/O modules and their applications
- Read a basic PLC wiring diagram and match terminal labels to physical wiring
- Understand cabinet infrastructure: fuses, DIN rail, grounding, and wire labeling

Module 2: Safety & Input/Output Devices

- Input devices (sensors, switches, safety devices) and output devices (motors, relays, valves) — how they work and how they connect to the PLC.
- Identify the main input device types and describe the signal each sends to the PLC
- Identify the main output device types and describe how the PLC drives each
- Understand the difference between standard I/O and safety I/O (dual-channel, OSSD)
- Recognize additional field devices: photoelectric sensors, encoders, solenoid valves, VFDs

Module 3: Software Setup & PLC Connectivity

- Installing PLC software, configuring IP addresses, and connecting to all three PLC platforms. Includes program backup/restore procedures.
- Configure a laptop IP address to communicate with a PLC on a given subnet
- Connect to an Omron NX1P2 using Sysmac Studio
- Connect to an Allen Bradley controller using Studio 5000
- Connect to a Productivity PLC using Productivity Suite
- Perform program upload (backup) and understand download procedures

Module 4: Reading & Troubleshooting Logic

- Using I/O dashboards, tag browsers, and ladder logic editors to trace signals and diagnose faults across all three platforms.
- Apply the 5-layer fault isolation model to systematically troubleshoot PLC issues
- Navigate the I/O dashboard to view real-time signal status
- Trace a signal from physical input through ladder logic to physical output
- Use forcing and monitoring tools safely and effectively
- Understand data types, tag structure, and variable scoping

Module 5: Area Scanner Programming

- Programming and verifying the Omron OS32C area scanner — defining safety zones, configuring OSSD outputs, and physical verification.
- Program or verify the configuration of an Omron OS32C area scanner
- Define protective and warning zones using the OS32C Configuration Tool
- Confirm OSSD outputs are functioning correctly via PLC monitoring
- Physically verify zone boundaries with measurement

Module 6: HMI Navigation & Fault Clearing

- Operating the HMI screen — login, alarm management, I/O status monitoring, and fault clearing procedures.
- Log into an HMI screen and navigate between key screens
- View and interpret active alarms and fault history
- Clear faults and reset the machine after a safe condition is restored
- Use the I/O status screen to verify field device communication
- Manually jog or operate outputs for maintenance testing

Module 7: Scenario-Based Fault Exercises

- Five realistic fault scenarios pre-staged on the bench. Diagnose and resolve each independently using the skills from Modules 1–6.
- Independently diagnose and resolve a tripped area scanner fault
- Identify and correct an IP connectivity failure
- Trace a missing sensor signal from PLC to field device
- Identify a logic fault (incorrect timer preset) by tracing the ladder rung
- Replace a faulty push button and verify proper operation

Module 8: Advanced Manufacturing Specifics

- Custom machine documentation, ladder logic building blocks, servo motor configuration, global variables, and adding I/O components to a PLC.
- Locate and use custom build documentation for Vermeer machines
- Understand core ladder logic instructions and their symbols
- Configure basic servo motor parameters in Sysmac Studio
- Create and use global variables with proper naming conventions
- Add I/O modules and devices to an existing PLC configuration